

| Question |     |  |  | Marking details                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Marks available |     |     |       |       |      |
|----------|-----|--|--|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-----|-----|-------|-------|------|
|          |     |  |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | AO1             | AO2 | AO3 | Total | Maths | Prac |
| 4        | (a) |  |  | <p><b>Indicative content:</b></p> <ol style="list-style-type: none"><li>1. Set up a pointer in the clamp stand and adjust its height.</li><li>2. The height (distance 2) should be at least 100 cm above the floor. Measure and record this height using a metre ruler.</li><li>3. Drop a single cake case from a height (distance 1) at least 15 cm above the pointer.</li><li>4. Using a stopwatch, measure and record the time it takes to fall from the pointer to the floor.</li><li>5. Repeat steps 3 and 4.</li><li>6. Repeat by adding more cake cases to the stack.</li></ol> <p><b>5–6 marks</b><br/>Detailed description of apparatus set-up and suitable values <b>and</b> the measurements taken.<br/><i>There is a sustained line of reasoning which is coherent, relevant, substantiated and logically structured. The candidate uses appropriate scientific terminology and accurate spelling, punctuation and grammar.</i></p> <p><b>3–4 marks</b><br/>Detailed description of either apparatus set-up and suitable values or the measurements taken <b>OR</b> limited description of both.<br/><i>There is a line of reasoning which is partially coherent, largely relevant, supported by some evidence and with some structure. The candidate uses mainly appropriate scientific terminology and some accurate spelling, punctuation and grammar.</i></p> | 6               |     |     | 6     |       | 6    |